KASUN JAYAMAHA *(Associate Software Engineer)*

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LinkedIn | **GitHub** | **Portfolio**

PERSONAL STATEMENT

Results-driven Associate Software Engineer with a solid academic foundation in Software Engineering and hands-on industry experience building **full-stack applications**, **scalable backend systems**, **AI-integrated workflows**, and **Industrial IoT (IIoT) solutions**. Specializing in **JavaScript/TypeScript** ecosystems **(Node.js, React, NestJS)**, cloud infrastructure (**AWS Bedrock, SQS, RDS, CloudFront, MediaConvert, IoT Core, Lambda, DynamoDB**), and edge-computing architectures (**ESP32/ESP-NOW**). Proven track record of delivering fault-tolerant distributed systems and publishing industry-tested research deployed in real-world manufacturing environments.

TECHNICAL SKILLS

- **Languages**: JavaScript, TypeScript, Python, C/C++, Java
- **Web & Frameworks**: React, Next.js, Node.js, Express, NestJS, Tailwind CSS
- **Mobile & IoT**: ESP32 (ESP-NOW, I2C, SPI), Embedded C/C++, Flutter, React Native, Industrial Sensors
- **Cloud & DevOps**: AWS (Bedrock, SQS, RDS, CloudFront, MediaConvert, IoT Core, Lambda, DynamoDB), Docker, Kubernetes, Git, CI/CD
- **Databases**: PostgreSQL, MongoDB, Firebase, Supabase, Prisma ORM
- **Architecture & Protocols**: Microservices, Edge Computing, MQTT, RESTful APIs, Agile/Scrum

EMPLOYMENT HISTORY

- **Associate Software Engineer**
  (Axceera (Pvt) Ltd, Sri Lanka)
  August 2026 – Present
    - Engineer full-stack web applications with modern frameworks, REST APIs, cloud workflows, and microservices architectures.
    - Collaborate in an Agile development environment to deliver high-performance features, code optimizations, and scalable cloud solutions across various AWS services.
    - Transitioned into a full-time associate role following the successful completion of a 1-year software engineering internship, demonstrating rapid technical growth and consistent project delivery.

- **Software Engineer Intern**
  (Axceera (Pvt) Ltd, Sri Lanka)
  August 2025 – August 2026
    - Contributed to production-level software development, backend API integrations, and database design for scalable web applications.

- Participated in sprint planning, code reviews, and bug resolution to ensure high software quality standards.
- Developed scalable frontend and backend modules utilizing modern web stack technologies to improve system efficiency.

MAJOR PROJECTS & RESEARCH

PredictiveOps – Industrial IIoT Predictive Maintenance System (Undergraduate Research)

- Architected an end-to-end IIoT monitoring and predictive maintenance framework targeting heavy industrial rotating machinery, successfully deployed on a Kiln Exhaust Blower Unit at Haycarb PLC.
- Engineered a decoupled dual-microcontroller gateway (ESP32) utilizing the connectionless ESP-NOW protocol to eliminate radio resource contention, achieving a 96.41% Packet Delivery Rate (PDR) in high electromagnetic interference environments.
- Designed an asynchronous, serverless cloud pipeline on AWS (IoT Core, Lambda, DynamoDB) executing 1g gravity vector isolation and Exponential Moving Average (EMA) smoothing under <300ms latency.
- Integrated a SIM800L cellular backhaul for non-IP autonomous SMS alerting during total facility network outages, achieving a 4.0-second alert dispatch time.

Quick Grubs – Food Delivery Management Platform

- Built a full-stack food delivery application using a microservices architecture for handling restaurant operations, delivery dispatching, and secure payment gateways.

PawPal – Pet Care and Adoption Platform

- Engineered a mobile-first application using Flutter designed to facilitate pet adoption, provide veterinary advice, and manage daily pet care routines.

Laundry Pro – Laundry Management System

- Built a Java-Servlets based system for managing laundry services, implementing core features for user management and comprehensive order tracking.

Nourish Lanka – Food Donation Management System

- Developed a donation-based platform designed to connect food donors with NGOs and underprivileged communities.

EDUCATION & QUALIFICATIONS

BSc (Hons) in Information Technology Specializing in Software Engineering

Sri Lanka Institute of Information Technology (SLIIT), Sri Lanka

Completed 2026 (Convocation expected September 2026)

- Undergraduate Thesis: AR-AI Guided Maintenance with IoT Monitoring for Industrial Machinery (PredictiveOps)

G.C.E. Advanced Level (English Medium)

Thurstan College, Colombo 07, Sri Lanka

- Results: 2C, 1S

G.C.E. Ordinary Level (English Medium)
St. Joseph's College, Nugegoda, Sri Lanka
- Results: 5A, 3B, 1S

CERTIFICATIONS & TECHNICAL WRITING

- AI/ML Engineer Stage-1 Certification – Centre for Open and Distance Education (CODE), SLIIT
- Technical Writer / Tech Blogger – https://medium.com/@ks2005755(Publishing insights on IoT, ESP32, embedded systems, and full-stack development)

REFERENCES

*References are available upon request.